EXERCISE 6.5

1. Find the parametric form of vector equation and Cartesian equations of a straight line passing through $(5,2,8)$ and is perpendicular to the straight lines $\vec{r} = (\hat{i} + \hat{j} - \hat{k}) + s(2\hat{i} - 2\hat{j} + \hat{k})$ and $\vec{r} = (2\hat{i} - \hat{j} - 3\hat{k}) + t(\hat{i} + 2\hat{j} + 2\hat{k})$.
2. Show that the lines $\vec{r} = (6\hat{i} + \hat{j} + 2\hat{k}) + s(\hat{i} + 2\hat{j} - 3\hat{k})$ and $\vec{r} = (3\hat{i} + 2\hat{j} - 2\hat{k}) + t(2\hat{i} + 4\hat{j} - 5\hat{k})$ are skew lines and hence find the shortest distance between them.
3. If the two lines $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-1}{4}$ and $\frac{x-3}{1} = \frac{y-m}{2} = z$ intersect at a point, find the value of m.
4. Show that the lines $\frac{x-3}{3} = \frac{y-3}{-1}, z-1=0$ and $\frac{x-6}{2} = \frac{z-1}{3}, y-2=0$ intersect. Also find the point of intersection.
5. Show that the straight lines $x+1 = 2y = -12z$ and $x = y+2 = 6z-6$ are skew and hence find the shortest distance between them.
6. Find the parametric form of vector equation of the straight line passing through $(-1,2,1)$ and parallel to the straight line $\vec{r} = (2\hat{i} + 3\hat{j} - \hat{k}) + t(\hat{i} - 2\hat{j} + \hat{k})$ and hence find the shortest distance between the lines.
7. Find the foot of the perpendicular drawn from the point $(5,4,2)$ to the line $\frac{x+1}{2} = \frac{y-3}{3} = \frac{z-1}{-1}$. Also, find the equation of the perpendicular.

6.8 Different forms of Equation of a plane

We have already seen the notion of a plane.

Definition 6.8

A vector which is perpendicular to a plane is called a **normal** to the plane.

Note

Every normal to a plane is perpendicular to every straight line lying on the plane.
A plane is uniquely fixed if any one of the following is given:

- a unit normal to the plane and the distance of the plane from the origin
- a point of the plane and a normal to the plane
- three non-collinear points of the plane
- a point of the plane and two non-parallel lines or non-parallel vectors which are parallel to the plane
- two distinct points of the plane and a straight line or non-zero vector parallel to the plane but not parallel to the line joining the two points.

Let us find the vector and Cartesian equations of planes using the above situations.

6.8.1 Equation of a plane when a normal to the plane and the distance of the plane from the origin are given

(a) Vector equation of a plane in normal form

Theorem 6.15

The equation of the plane at a distance p from the origin and perpendicular to the unit normal vector $\hat{d}$ is $\vec{r} \cdot \hat{d} = p$.

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